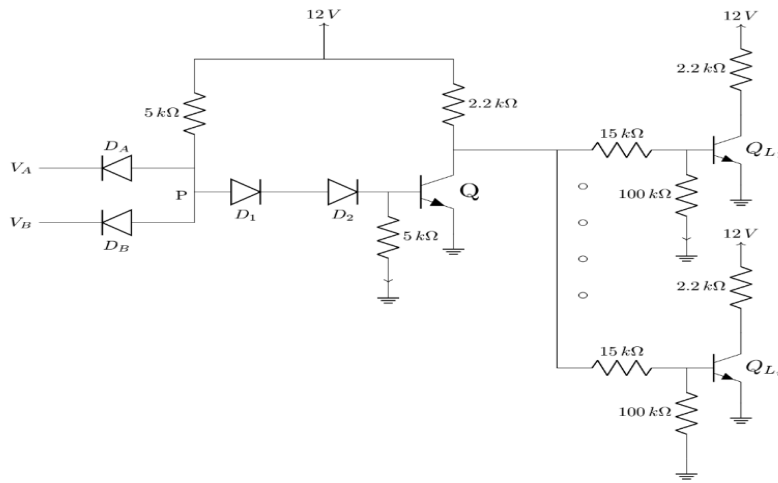


Part: A

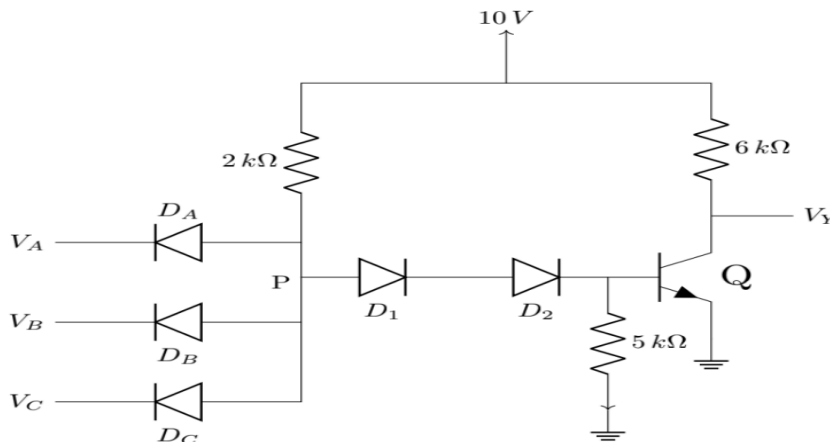
Question 1: For the given circuit in below, $V_{OH} = 10\text{ V}$, $V_{OL} = 0.2\text{ V}$, $\beta_F = 30$

- (a) When $N = 1$, what logic function do we get from the output of the load?
- (b) Find the maximum fanout.
- (c) For $N = \text{maximum fanout}$, find the maximum power dissipation in the loads.

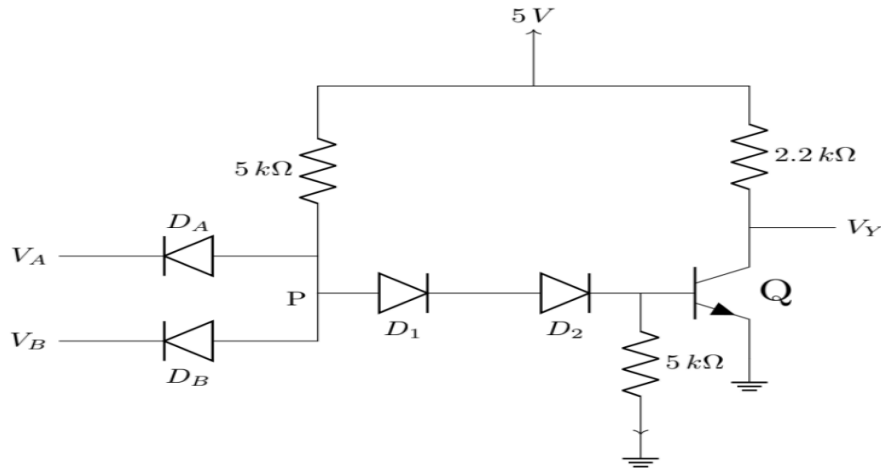


Question 2: For the bellow circuit, $V_{OH} = 9.5\text{ V}$, $V_{OL} = 0.1\text{ V}$ & $\beta_F = 30$.

- (a) Find the value of β_{Fmin} when all inputs are high.
- (b) Find the average power dissipation of the circuit.
- (c) Now suppose this same circuit is used as loads and connected to this driver circuit. Find the maximum fanout.



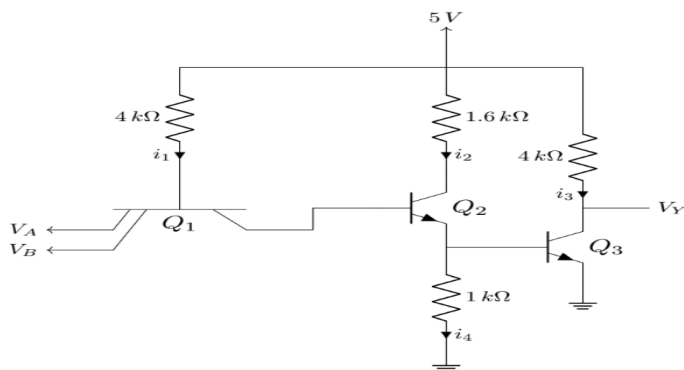
Question 3:



For the DTL NAND gate— find all voltage and currents for all logic cases with verification of assumption. Assume, high input is 5 V and low input is 0.2 V. Also, $\beta_F = 30$.

Part: B

Question 4:



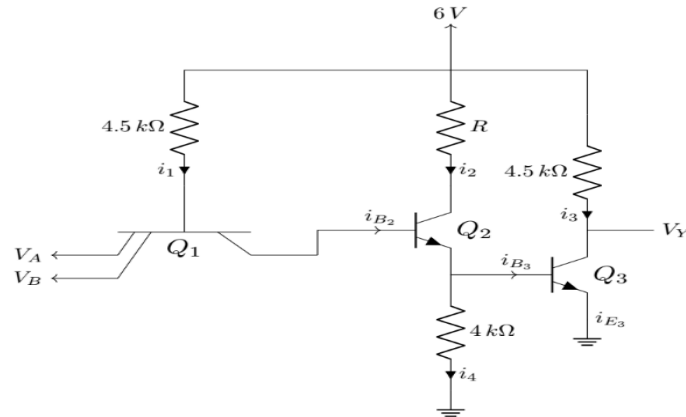
For the TTL NAND gate shown above, $\beta_F = 25$, $\beta_R = 0.1$, $V_{OH} = 3.4\text{ V}$, $V_{OL} = 0.2\text{ V}$.

Assume

high input = 5 V and low input = 0.2 V.

- If at least one input is low, find i_1 , i_2 , i_3 , i_4 .
- Find all the currents when both inputs are high.
- Find the maximum and average power dissipation of the circuit.

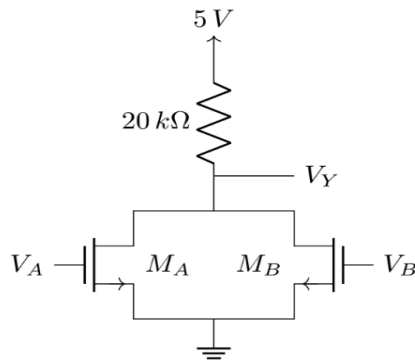
Question 5:



For the given circuit, $\beta R = 0.2$, $V_{CEsat} = 0.2 \text{ V}$. If $i_{E3} = 4.74 \text{ mA}$ when both inputs are high, find the value of R .

Part: C

Question 6:

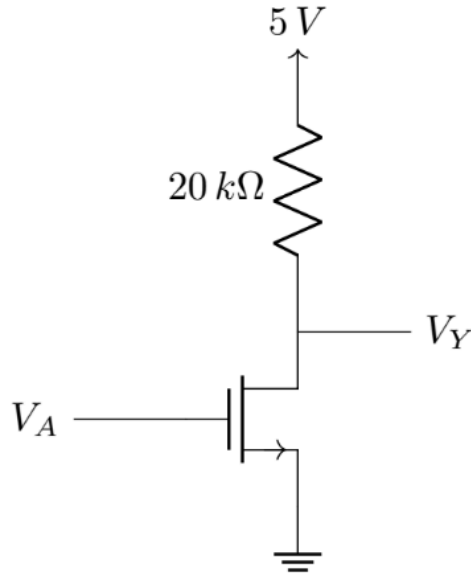


For the NOR gate shown, $V_{TN_A} = 0.25 \text{ V}$, $V_{TN_B} = 0.3 \text{ V}$, $K_{n_A} = 0.4 \frac{\text{A}}{\text{V}^2}$, $K_{n_B} = 0.5 \frac{\text{A}}{\text{V}^2}$.

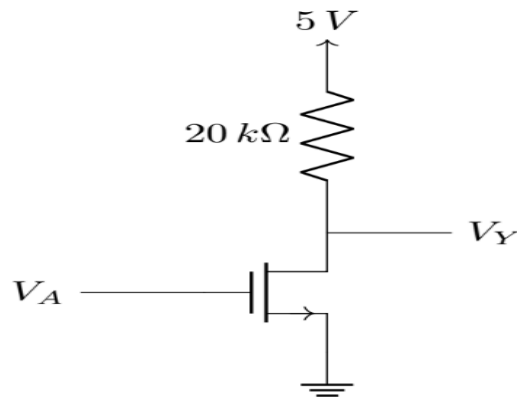
- Find the lower threshold of output voltage.
- Find the average power dissipation.
- If a new NMOS MC is used instead of MB such that the drain currents of MA & MC are equal when both inputs are high, find the value of K_{n_C} . It is given that $V_{TN_C} = V_{TN_B} = 0.3 \text{ V}$.

Question 7:

For the given NMOS inverter, $V_{TN} = 0.5 \text{ V}$, $K_n = 0.3 \frac{\text{mA}}{\text{V}^2}$. Find the output voltage V_Y for $V_A = 1.5 \text{ V}$ & 5 V .



Question 8: For the circuit in Question 7, find the maximum drain current, transition point from triode to saturation and the maximum power dissipation.

Question 9:

Parameters for the driver MD, $V_{TND} = 1 \text{ V}$, $K_{nD} = 50 \frac{\mu\text{A}}{\text{V}^2}$.

Parameters for the load, ML, $V_{TNL} = 1 \text{ V}$, $K_{nL} = 10 \frac{\mu\text{A}}{\text{V}^2}$.

Determine the output voltage and the current for both high and low input cases.

Question 10: Find the average power dissipation of the circuit in Question 9.

Question 11: Implement a two-input NOR gate using CMOS logic.

Question 12: Implement the logic function $Y = (AC+D)(BC+D)$ using CMOS logic.